

BACKGROUND

About Me

 **Masters in Computer Science**

Wichita State University (May 2024)

 Interested in the healthcare sector

 Loves cats, dogs & traveling

 Dream destination: Santorini, Greece

Externship Projects

Stanford Medical Process Design & Strategy Consulting

01 Stakeholder & Role Clarity

Mapped care team roles using real patient cases to surface inefficiencies.

02 Workflow Diagnostics

Root Cause Analysis (5 Whys, Fishbone) uncovering clinic pain points.

03 Clinical Decision Mapping

Translated KDIGO, KDOQI & CMS guidelines into Mermaid.js decision trees.

04 Decision-Support Dashboard

Prototyped a role-aware dashboard with Kotter's Change Model adoption plan.

Externship Project Summary

Improving Nephrology Clinic Operations

01 Stakeholder & Role Clarity

Mapped care team roles and scopes of practice using real patient cases to identify inefficiencies and their impact on patient outcomes.

02 Workflow Diagnostics

Investigated broken workflows using Root Cause Analysis (5 Whys, Fishbone), uncovering internal clinic pain points and exploring AI-supported solutions.

03 Clinical Decision Mapping

Translated KDIGO, KDOQI, and CMS guidelines into decision trees to improve escalation and triage. Visualized workflows using Mermaid.js and Claude AI.

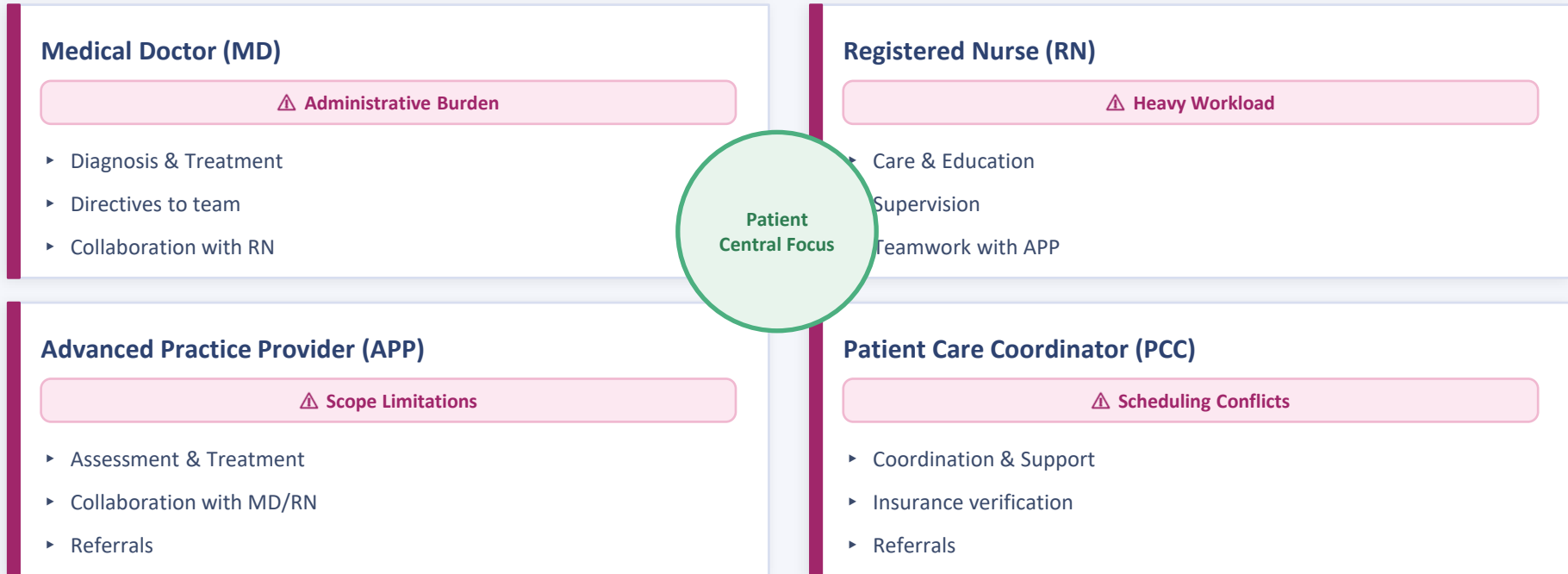
04 Decision-Support Dashboard

Prototyped an interactive dashboard tailored to clinical roles. Applied Kotter's Change Model to plan for real-world adoption and impact.

Project 1 Deliverables

Stakeholder & Role Clarity

Mapped key stakeholders in nephrology clinics to understand roles, responsibilities, and care coordination.



Project 2 Deliverables

Workflow Diagnostics — Root Cause Analysis

 **PMI: Elevated potassium level (6.2 mmol/L) not addressed — resulted in delayed intervention and patient harm risk**

Fishbone (Ishikawa) Diagram — Domains

People

- Role ambiguity on escalation responsibility
- No accountability for lab follow-up

Process

- No standardized escalation protocol
- Manual alert tracking prone to misses

Technology

- EHR alerts not configured to notify APP/MD
- No tool to track alert acknowledgment

Environment

- High patient volume limits bandwidth
- Busy clinic hours reduce oversight

5 Whys Analysis

Why 1: Why not addressed?

EHR alert flagged by RN but not escalated to APP/MD.

Why 2: Why not escalated?

No clear or enforced escalation protocol for RNs.

Why 3: Why no protocol?

Clinic lacked SOPs and training for critical lab handoffs.

Why 4: Why no SOPs?

Leadership hadn't prioritized lab alert workflows.

Why 5: Why not a priority?

Risk underestimated; no data or feedback loop in place.

Root Cause: Missing structured escalation protocol & feedback loop

Project 3 Deliverable

Clinical Decision Mapping

 **PMI: Elevated potassium level (6.2 mmol/L) not addressed — resulted in delayed intervention and patient harm risk**

01

Text-Based Decision Tree

Drafted clinical decision logic for potassium level escalation — defining urgency tiers, role responsibilities, and timing thresholds for each severity band.

Text Logic

02

Mermaid.js Flowchart

Translated the decision logic into a digitized visual flowchart using Mermaid.js, producing a format usable during time-constrained clinical decision-making.

Mermaid.js + Claude AI

03

Guideline Mapping

Grounded every escalation threshold in KDIGO, KDOQI, and CMS clinical guidelines to ensure the decision tree reflects established nephrology standards.

KDIGO · KDOQI · CMS

Outcome: A role-aware, guideline-driven escalation flowchart enabling fast, consistent clinical decision-making under time pressure.

Project 4 Deliverables

Decision-Support Dashboard & Change Management Plan

 **PMI: Elevated potassium level (6.2 mmol/L) not addressed — resulted in delayed intervention and patient harm risk**

Training Strategy

Role-Based Training Sessions

15–20 min role-specific demos for RNs, APPs, Nephrologists & PCCs

Live Patient Scenarios

Real past cases (PHI removed) to walk through dashboard decision-making

Quick-Reference Guides

Laminated 1-pagers and badge cards for when and how to use the tool

Peer Champions

1–2 super users per team to support peers in the first 2 weeks post-launch

Tool Embedded in Workflow

Dashboard linked in clinic bookmarks and internal SOP documents

Resistance Mitigation Plan

Scope creep fears

→ Reinforce dashboard shows only role-appropriate actions

Staff confusion or bugs

→ Feedback form or Slack channel for real-time reporting

Low clinical buy-in

→ Clinical champion endorsement from a respected nephrologist

Workflow disruption

→ Start with 1–2 teams; allow optional use initially

Project 4 Deliverables — Continued

Decision-Support Dashboard Prototype & Success Metrics

Dashboard Prototype



Created a role-aware decision-support dashboard prototype for the externship project.



Prototype code hosted on GitHub Repository: Stanford Nephrology Dashboard.



Live prototype: Stanford Nephrology Decision-Support Dashboard Prototype (Final).

Success Metrics



Usage Rate

% of staff using dashboard within first 30 days



Escalation Timeliness

Reduction in time from critical K+ result to provider intervention



Error Reduction

Decrease in missed labs or inappropriate escalations



Staff Satisfaction

Post-rollout survey on tool usability and clarity



Adherence

% of escalations following protocol steps shown in the tool

Insights Gained

Reflections, Concerns & Next Steps

Food for Thought

- ▶ How do we ensure lab results are reviewed quickly during busy clinic hours?
- ▶ Can we introduce alerts without overwhelming already-stretched clinical staff?

Concerns Raised

- ▶ Too many alerts may cause staff to ignore them (alert fatigue)
- ▶ New tools might slow down existing workflows during the learning curve
- ▶ Some team members may need additional support learning the dashboard

What's Next

- ▶ Collect feedback from a controlled test run with 1–2 clinic teams
- ▶ Observe staff usage patterns and iterate on design accordingly
- ▶ Work with IT to formally link the dashboard to the clinic's EHR systems

ACKNOWLEDGEMENTS

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 Stanford Medicine Lead

Kelly Chen

Provided real-world insights into nephrology operations and guided the project from start to finish.

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Satya Yagnambhat · Carl Joel Tilos · Mike Eng

Shaped each project deliverable with thoughtful advice and hosted weekly collaborative sessions.

 Peers & Collaborators

Anish · Devan · Keoni · and cohort

Contributed teamwork, peer feedback, and support throughout the externship.